

A Mixture Relaxation Model for Thermal Nonequilibrium in Convective Flow Boiling

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ABSTRACT

Convective flow boiling is a key process in Light Water Reactors (LWRs), where accurate prediction of thermal nonequilibrium between the phases is essential for both performance evaluation and safety assessment. Traditional two-fluid models offer detailed representations of the involved mechanisms but rely on complex interfacial area and heat/mass transfer closures. Developed as an alternative approach applicable to one-dimensional high pressure two-phase flows, this work introduces a Mixture Relaxation Model (MRM) that provides a simplified yet physically consistent representation of interfacial heat and mass transfer through a time-relaxation framework.

The model solves mixture mass, momentum, and energy conservation equations, complemented by vapor-phase mass and energy, allowing both phases to depart from thermal equilibrium while maintaining a simplified hydrodynamic coupling. The formulation replaces explicit interfacial area and interfacial heat transfer models with a lumped relaxation-based treatment of interfacial exchanges, reducing complexity while preserving essential physics. Required closure relations include wall heat flux partitioning, wall heat transfer models, and interfacial relaxation times derived from physically based formulations.

Implemented in the open-source OpenSTREAM code, the MRM is applied to two reference datasets relevant to subcooled boiling and post-boiling transition evaluations. The results confirm that the model reproduces correct conceptual trends over the selected conditions and accurately predicts void fraction, wall temperature, and the expected relaxation toward equilibrium along adiabatic sections. Hence, the model captures key thermal nonequilibrium interfacial phenomena using a remarkably simple set of governing equations. This work provides a robust and computationally efficient alternative to traditional two-fluid models, offering strong potential for integration into system and subchannel codes for high pressure LWR safety analysis.

KEYWORDS

TWO-PHASE FLOW, RELAXATION, SUBCOOLED BOILING, POST-BOILING TRANSITION.

1. INTRODUCTION

Accurate prediction of convective flow boiling is essential for thermal-hydraulic analyses in systems where two-phase flows evolve under strong wall-driven thermal nonequilibrium, i.e., subcooled boiling and post-boiling transition. Traditional modeling approaches often rely on mixture formulations (e.g., Homogeneous Equilibrium Model) or on full two-fluid frameworks, but these can be either overly restrictive (e.g., by assuming instantaneous thermal equilibrium) or complex and sometimes unreliable due to their dependence on complex interfacial area and interfacial transfer models. To address this gap, this work introduces a Mixture Relaxation Model (MRM): a one-dimensional, five-equation formulation that captures phase thermal nonequilibrium while retaining the simplicity and robustness of mixture-based approaches.

The proposed MRM extends concepts used in classical Homogeneous Relaxation Models (HRM), traditionally applied to flashing or depressurization flows [1][2][3], and more seldomly to post-boiling transition [4] and subcooled boiling [5][6]. A unified approach (flashing and post-boiling transition) has also been introduced in [7]. By incorporating time-relaxed interfacial heat and mass transfer driven explicitly by wall heat input, the proposed model enables consistent modeling of subcooled boiling, saturated boiling, and post-boiling transition using a unified mathematical framework. Both liquid and vapor phases may depart from thermal equilibrium, while hydrodynamic coupling remains simplified (an appropriate assumption for high-pressure water flows where phase slip remains limited).

The model is particularly relevant for Light Water Reactor (LWR) applications. Under normal operating conditions, subcooled boiling influences coolant density and thus reactor moderation, power distribution, and the margin to Critical Heat Flux. Under Anticipated Operational Occurrences (AOOs), the ability to predict thermal nonequilibrium in the post-boiling transition region is crucial for evaluating time-at-temperature fuel safety criteria, currently under consideration [8]. By relying on a time-relaxation approach, the MRM captures these effects without requiring detailed interfacial area transport and interfacial heat transfer models, enabling efficient simulation across a broad range of boiling regimes.

This paper presents the governing mixture and vapor-phase conservation equations together with the relaxation-based closure relations that form the basis of the MRM. Implemented in the mixture solver of the open-source OpenSTREAM two-phase flow computational environment [9][10], the model's phenomenological behavior is examined, including the physical interpretation of the calculated relaxation time in both subcooled boiling and post-boiling transition regimes. Further validations against representative experimental datasets demonstrate the MRM capabilities to simulate thermal nonequilibrium in convective flow boiling.

NOMENCLATURE

A	Cross-sectional area	x	Vapor mass quality	<u>Subscript</u>	
a_i	Volumetric interfacial area	z	Longitudinal position	do	Film dryout
d	Diameter			g	Vapor phase
g	Gravitational acceleration			h	Hydraulic
h	Specific enthalpy	<u>Greek</u>		i	Interfacial
htc	Heat transfer coefficient	α	Void fraction	l	Liquid phase
k	Thermal conductivity	Γ	Evaporation mass flux	$relax$	Relaxation
p	Pressure	θ	Inclination angle	s	Saturated
q''	Heat flux	κ	Thermal diffusivity	w	Wall
T	Temperature	Λ	Condensation mass flux	<u>Superscript</u>	
t	Time	Π	Perimeter	eq	Thermodynamic equilibrium
u	Velocity	ρ	Density	$+$	Advection
W	Mass flow rate	τ	Interfacial stress		

2. TWO-PHASE CONSERVATION EQUATIONS

This section presents the governing conservation equations forming the foundation of the Mixture Relaxation Model (MRM). The model is constructed by combining mixture-level mass, momentum, and energy equations with additional vapor-phase conservation laws that enable explicit treatment of phase thermal nonequilibrium using a time relaxation approach. Derived from time-averaging and cross-section-averaging of the instantaneous conservation laws [11], these equations represent a simplified framework that is suitable for modeling subcooled boiling, saturated boiling, and post-boiling transition conditions. The closure relations required to complete the formulation are also introduced, with the exception of the interfacial relaxation-time models, which are addressed separately in Section 3.

2.1. Mixture

This subsection summarizes the one-dimensional, time-dependent conservation equations governing mass, momentum, and energy transport in the two-phase mixture. Derived from the summation of the individual phase conservation laws, they describe the global behavior of the system and constitute the backbone of the MRM.

The mixture mass conservation equation is:

$$\frac{\partial}{\partial t} \left(\frac{W}{u} \right) + \frac{\partial W}{\partial z} = 0 \quad (1)$$

where $u (= W/(\rho \cdot A))$ is the mass-centered mixture velocity (static mass-weighted mean).

For a straight channel with no obstruction, the mixture momentum conservation equation is:

$$\frac{\partial W}{\partial t} + \frac{\partial}{\partial z} (u^+ \cdot W) = -A \cdot \left(\frac{\partial p}{\partial z} + \cos \theta \cdot g \cdot \rho \right) - \Pi_w \cdot \tau_w \quad (2)$$

where u^+ is the mass flux-centered (advection) mixture velocity.

Neglecting kinetic energy, frictional heating, transient pressure work, and other minor contributions, the mixture energy conservation equation becomes:

$$\frac{\partial}{\partial t} \left(h \cdot \frac{W}{u} \right) + \frac{\partial}{\partial z} (h^+ \cdot W) = \Pi_w \cdot q_w'' \quad (3)$$

where h is the mass-centered mixture enthalpy and h^+ is the mass flux-centered (advection) mixture enthalpy.

This set of mixture equations allows inhomogeneous phase velocities through selected slip or drift flux models (see Section 2.3.1). Under equal phase velocity assumptions, $u = u^+$ and $h = h^+$, reducing the formulation to a Homogenous Relaxation Model (HRM).

2.2. Vapor phase

To account for thermal nonequilibrium between phases, the mixture equations are complemented by one-dimensional, time-dependent, conservation equations for vapor mass and vapor energy. These equations are formulated to explicitly represent thermal nonequilibrium processes through interfacial evaporation and condensation. Their solution provides the vapor mass flow rate and vapor enthalpy, allowing a physically consistent determination of phase mass quality and phase temperature.

The vapor mass conservation equation is:

$$\frac{\partial}{\partial t} \left(\frac{W_g}{u_g} \right) + \frac{\partial W_g}{\partial z} = A \cdot a_i \cdot (\Gamma_i - \Lambda_i) + \Pi_w \cdot \Gamma_w \quad (4)$$

Assuming phase change at the wall (Γ_w) and interfaces (Γ_i , Λ_i) occurs at the respective phase enthalpies, the vapor energy conservation equation becomes (after subtraction by Equation (4) multiplied by h_g):

$$\frac{W_g}{u_g} \cdot \frac{\partial h_g}{\partial t} + W_g \cdot \frac{\partial h_g}{\partial z} = A \cdot a_i \cdot (h_l - h_g) \cdot \Gamma_i + \Pi_w \cdot q_g'' \quad (5)$$

The fundamental assumption of the MRM is to consider that the interfacial mass exchanges are governed by the deviation from thermal equilibrium under a time relaxation principle:

$$A \cdot a_i \cdot (\Gamma_i - \Lambda_i) = \frac{x^{eq} \cdot W - W_g}{u_g \cdot t_{Relax}} \quad (6)$$

Assuming that interfacial evaporation and condensation cannot occur simultaneously (within the same computational node), the individual interfacial mass transfer rates become:

$$\text{Interfacial evaporation:} \quad A \cdot a_i \cdot \Gamma_i = \max \left(\frac{x^{eq} \cdot W - W_g}{u_g \cdot t_{relax}}, 0 \right) \quad (7)$$

$$\text{Interfacial condensation:} \quad -A \cdot a_i \cdot \Lambda_i = \min \left(\frac{x^{eq} \cdot W - W_g}{u_g \cdot t_{relax}}, 0 \right) \quad (8)$$

where the relaxation time, t_{relax} , may differ between evaporation and condensation processes.

Substituting these relations into Equations (4) and (5), the vapor conservation equations become:

$$\frac{\partial}{\partial t} \left(\frac{W_g}{u_g} \right) + \frac{\partial W_g}{\partial z} = \frac{x^{eq} \cdot W - W_g}{u_g \cdot t_{relax}} + \Pi_w \cdot \Gamma_w \quad (9)$$

and

$$\frac{W_g}{u_g} \cdot \frac{\partial h_g}{\partial t} + W_g \cdot \frac{\partial h_g}{\partial z} = \max \left(\frac{x^{eq} \cdot W - W_g}{u_g \cdot t_{relax}}, 0 \right) \cdot (h_l - h_g) + \Pi_w \cdot q_g'' \quad (10)$$

The complex volumetric interfacial area and interfacial mass transfer terms are hence replaced by a single representative time scale governing relaxation toward thermal equilibrium. Once the vapor mass flow rate and vapor enthalpy are resolved, the vapor mass quality ($x = W_g/W$) and vapor temperature (T_g , obtained from an equation of state based on h_g and p) can be readily calculated.

2.3. Closure laws

This subsection introduces the closure laws required to complete the mixture and vapor conservation equations. These include interfacial shear stress models, wall heat-flux partitioning, and wall friction and heat transfer models. The closure for the interfacial mass transfer based on time-relaxation is addressed separately in Section 3.

2.3.1. Interfacial shear stress

Although interfacial shear does not appear explicitly in the MRM governing equations, it is implicitly considered in the void fraction model and affects mixture density, velocity, and enthalpy. The MRM can accommodate void fraction models based on homogeneous assumption, velocity slip or any drift flux model.

2.3.2. Wall heat flux partitioning

Wall heat flux partitioning specifies how the applied wall heat flux is distributed between liquid heating, evaporation, and vapor heating. For simplicity, a linearly increasing liquid heating/evaporation scheme is applied in the subcooled region (between $-0.2 \leq x^{eq} \leq 0$), while the wall heat flux is directed entirely to the vapor-phase beyond the post-boiling transition (i.e., potential heat transfer to liquid is neglected). The results are not excessively sensitive to the selected subcooled boiling x^{eq} boundaries. However, improvements may be considered in the future, supported by further experimental motivations.

2.3.3. Wall friction and wall heat transfer

Wall friction is modeled using a relevant set of single-phase liquid correlation (e.g., Blasius) and two-phase flow friction multiplier. For the relatively high pressure, steady-state validation cases considered in this paper, this choice has negligible influence on the results and is therefore not discussed further.

Convective wall heat transfer is modeled using the Thom correlation [12] upstream of the boiling transition and the Dittus-Boelter correlation [13] (to superheated vapor) downstream, such that:

$$T_w = T_g + \frac{q_w''}{htc_w} \quad (11)$$

with

$$Nu = \frac{htc_w \cdot d_h}{k_g} = 0.023 \cdot Re_g^{0.8} \cdot Pr_g^{0.4} \quad (12)$$

Note that all vapor properties are calculated at the local (i.e., superheated) temperature.

2.4. Numerical implementation

The MRM is implemented in the mixture solver of the OpenSTREAM computational environment [9][10]. The governing conservation equations form a set of first-order advection equations with nonlinear source terms. These are solved using a simple first-order upwind scheme, which is sufficient for this purpose and consistent across all available solvers. Phase thermophysical properties are computed using the CoolProp library [14].

Time integration is performed using a fully implicit backward Euler discretization. Nonlinear closure terms and field couplings are handled using a fixed-point iteration method at each timestep until convergence is achieved. This approach ensures numerical stability without imposing any timestep restrictions, resulting in a robust solution algorithm.

The transient solver is used for both steady-state and transient simulations. The steady-state solution is obtained by solving a pseudo-transient with large time steps (e.g., 1 second) to rapidly reach convergence.

3. INTERFACIAL RELAXATION TIME

This section presents the development of the closure laws related to the interfacial time-relaxation processes that enable the Mixture Relaxation Model (MRM) to capture thermal nonequilibrium in convective boiling flows. This relaxation time encapsulates the combined effects of volumetric interfacial area and interfacial heat and mass transfer rates into a single, physically interpretable, parameter. Although t_{relax} is expected to vary along the flow path as interfacial topology evolves, these variations are expected to remain smooth in well-coupled two-phase flows. This is consistent with the experimental observations reported in [15], where high-pressure boiling flow structures were found to evolve continuously over a wide range of increasing power, without any indication of abrupt transitions or discontinuities.

Starting from the general expression obtained from the vapor conservation law (Equation (6)), the relaxation time governing interfacial mass transfer can be expressed as follows:

$$t_{Relax} = \frac{x^{eq} \cdot W - W_g}{u_g \cdot A \cdot a_i \cdot (\Gamma_i - \Lambda_i)} = \rho \cdot \frac{u}{u_g} \cdot \frac{x^{eq} - x}{a_i \cdot (\Gamma_i - \Lambda_i)} = \rho_g \cdot \frac{\alpha}{x} \cdot \frac{x^{eq} - x}{a_i \cdot (\Gamma_i - \Lambda_i)} \quad (13)$$

where the deviation from thermal equilibrium can be developed and further expressed in terms of phase temperature deviation from saturation:

$$x^{eq} - x = \frac{x \cdot C_{p,g} \cdot (T_g - T_s) + (1 - x) \cdot C_{p,l} \cdot (T_l - T_s)}{h_{lg}} \quad (14)$$

In the following sub-sections, the relaxation time formulation is adapted to the distinct interfacial transfer mechanisms active in subcooled boiling (Section 3.1) and in the post-boiling transition region (Section 3.2), under reasonable assumptions. This leads to practical expressions relating t_{relax} to (mostly) readily available thermal-hydraulic quantities.

3.1. Subcooled boiling

This subsection focuses on the relaxation mechanisms governing interfacial condensation in the subcooled boiling region, which corresponds to the region where vapor is generated at the wall (due to evaporation from wall heat flux) while the liquid phase remains subcooled. The formulation is derived based on the same principles as in [5][6], where continuous subcooled liquid drives the interfacial condensation process. The dispersed saturated vapor phase is assumed saturated ($T_g = T_s$).

Considering only interfacial condensation ($\Gamma_i = 0$), Equation (13) reduces to:

$$t_{relax} = \rho_g \cdot \frac{\alpha}{x} \cdot \frac{x^{eq} - x}{-a_i \cdot \Lambda_i} \quad (15)$$

The interfacial condensation rate and associated interfacial heat transfer coefficient are expressed as:

$$-\Lambda_i = \frac{htc_i \cdot (T_l - T_s)}{h_{lg}} \quad htc_i = Nu_i \cdot \frac{k_l}{d_g} \quad (16)$$

Assuming a dispersed spherical vapor phase, the volumetric interfacial area becomes:

$$a_i = \frac{6 \cdot \alpha}{d_g} \quad (17)$$

Combining Equations (15)-(17) together with Equation (14) and $T_g = T_s$, a non-dimensional relaxation time for interfacial condensation is obtained:

$$t_{relax}^* = \frac{\kappa_l \cdot t_{relax}}{d_h^2} = \frac{\rho_g}{\rho_l} \cdot \frac{1-x}{x} \cdot \frac{1}{6 \cdot Nu_i} \cdot \left(\frac{d_g}{d_h}\right)^2 \quad (18)$$

This quantity is analog to a Fourier number, which traditionally represents the ratio of the rate of heat diffusion to the rate of thermal energy storage within a material, commonly used in transient heat conduction analysis. In the current context, however, it compares the rate of heat diffusion in the continuous phase to the relaxation time governing interfacial energy exchanges. As such, it represents a dimensionless measure of the thermal response time for interfacial mass transfer under nonequilibrium conditions.

Under homogeneous flow assumption, the interfacial thermal resistance is modeled as pure diffusion (i.e., $Nu_i = 2$), representing a conduction-limited interface without convective enhancement. This simplification avoids introducing additional uncertain closure models (phase velocity slip and interfacial Nusselt number correlation) and confines any required model calibration to a single parameter (see below), consistent with the lumped relaxation time formulation.

The only remaining unknown is the local vapor (Sauter mean) diameter, d_g . For the applications presented in Section 4.1, a calibrated constant value will be used. Although this characteristic length is expected to be of the same order as the actual bubble diameter, it should not be interpreted directly as such. Rather, it represents an effective length scale for the dispersed phase that incorporates all relevant corrections associated with the modeling assumptions and simplifications.

The resulting expressions capture how relaxation time decreases with, e.g., smaller bubbles and higher liquid thermal diffusivity, consistent with faster interfacial condensation and reduced nonequilibrium. In addition, as $x \rightarrow 0$, the relaxation time is expected to be very large, allowing the initiation of wall boiling in the subcooled region.

3.2. Post-boiling transition

This subsection develops the relaxation-time closure for interfacial evaporation in the post-boiling transition region, which corresponds to the region downstream the wall boiling transition characterized by superheated vapor (due to direct heat input from wall heat flux) while the liquid phase remains saturated. The formulation is derived based on the same principles as in [4], assuming Dispersed Film Flow Boiling (DFFB), where continuous superheated vapor drives the interfacial evaporation process. The dispersed saturated liquid droplets are assumed saturated ($T_l = T_s$). The derivation mirrors the approach used for subcooled boiling but with the roles of the phases reversed, and it incorporates the evolution of local droplet size (Sauter mean diameter) along the post-boiling transition region.

Considering only interfacial evaporation ($\Lambda_i = 0$), Equation (13) reduces to:

$$t_{relax} = \rho_g \cdot \frac{\alpha}{x} \cdot \frac{x^{eq} - x}{a_i \cdot \Gamma_i} \quad (19)$$

The interfacial evaporation and associated interfacial heat transfer coefficient are expressed as:

$$\Gamma_i = \frac{htc_i \cdot (T_g - T_s)}{h_{lg}} \quad htc_i = Nu_i \cdot \frac{k_g}{d_l} \quad (20)$$

Assuming a dispersed spherical liquid phase, the volumetric interfacial area becomes:

$$a_i = \frac{6 \cdot (1 - \alpha)}{d_l} \quad (21)$$

Combining Equations (19)-(21) together with Equation (14) and $T_l = T_s$, a non-dimensional relaxation time (similar to Equation (18)) for interfacial evaporation is obtained:

$$t_{relax}^* = \frac{\kappa_g \cdot t_{relax}}{d_h^2} = \frac{\alpha}{1 - \alpha} \cdot \frac{1}{6 \cdot Nu_i} \cdot \left(\frac{d_l}{d_h}\right)^2 \quad (22)$$

Under homogeneous flow assumption, the interfacial thermal resistance is again simplified using the conduction-limited value ($Nu_i = 2$).

The only remaining unknown is the local liquid (Sauter mean) droplet diameter, d_l . Within the DFFB flow regime, a reasonable simplification is to assume constant droplet number density (i.e., no coalescence or breakup) [16] downstream from the boiling transition (film dryout, do), which implies:

$$\frac{1 - \alpha}{d_l^3} = \frac{1 - \alpha_{do}}{d_{l,do}^3} \quad (23)$$

Hence, the relaxation time can be expressed in terms of the droplet Sauter mean diameter set at the boiling transition (see application example in Section 4.2), instead of the axially varying local downstream value:

$$t_{relax}^* = \frac{\kappa_g \cdot t_{relax}}{d_h^2} = \frac{\alpha}{(1 - \alpha)^{1/3}} \cdot \frac{1}{12} \cdot \frac{1}{(1 - \alpha_{do})^{2/3}} \cdot \left(\frac{d_{l,do}}{d_h}\right)^2 \quad (24)$$

The resulting expressions capture how relaxation time decreases with, e.g., smaller droplets and higher vapor thermal diffusivity, consistent with faster interfacial evaporation and reduced nonequilibrium. For the applications presented in Section 4.2, a calibrated constant value will be used for $d_{l,do}$. This represents an effective length scale for the dispersed phase that incorporates all relevant corrections associated with the modeling assumptions and simplifications.

4. PHENOMENOLOGICAL BEHAVIOR AND VALIDATION

A concise demonstration of the Mixture Relaxation Model (MRM) is presented under representative boiling conditions. The model's ability to capture thermodynamic nonequilibrium is illustrated for both subcooled boiling and post-boiling transition, highlighting the predicted evolution of phase mass flow rates, void fraction, vapor superheating, and wall temperature. Comparisons against selected experimental datasets confirm that the MRM reproduces the expected physical trends and remains bounded between the thermal-equilibrium and frozen-flow models.

4.1. Subcooled boiling

This section examines subcooled boiling simulations. Rather than relying on a subcooled boiling correlation (e.g., [17]) to calculate the vapor mass quality ($x > x^{eq}$) from which the void fraction is derived, the MRM uses the time-relaxed vapor conservation equations (Section 2.2) along with a wall boiling partitioning model (Section 2.3.2).

Two test cases from the RELAP5/MOD3 validation database [18] have been selected to illustrate the MRM's capabilities. Void fraction in uniformly heated convective flow boiling was measured using gamma-ray radiography in water at relatively high pressure [19][20]. The selected boundary conditions are listed in Table 1. In the second case, the heated length is followed by an adiabatic section, allowing to observe vapor condensation (the mixture being near the saturation limit at end of heated length).

Table 1: Selected subcooled boiling test cases

Test #	Reference	Diameter [mm]	Length [m]	Pressure [MPa]	Mass flux [kg/s/m ²]	T _{inlet} [C]	Heat flux [kW/m ²]
1	[19]	12	1.5 (heated)	4.4	994	190	900
2	[20]	12	1.0 (heated) 0.4 (non-heated)	6.9	980	231	824

For demonstration purposes, a calibrated constant d_g value (≈ 0.32 mm) is used to calculate the relaxation time related to the interfacial condensation (according to Equation (18) with $Nu_i = 2$). A simple drift flux void fraction model [21] is adopted. Figure 1 and Figure 2 show the axial distributions of wall heat flux partitioning, relaxation time, mass quality and void fraction together with the experimental data.

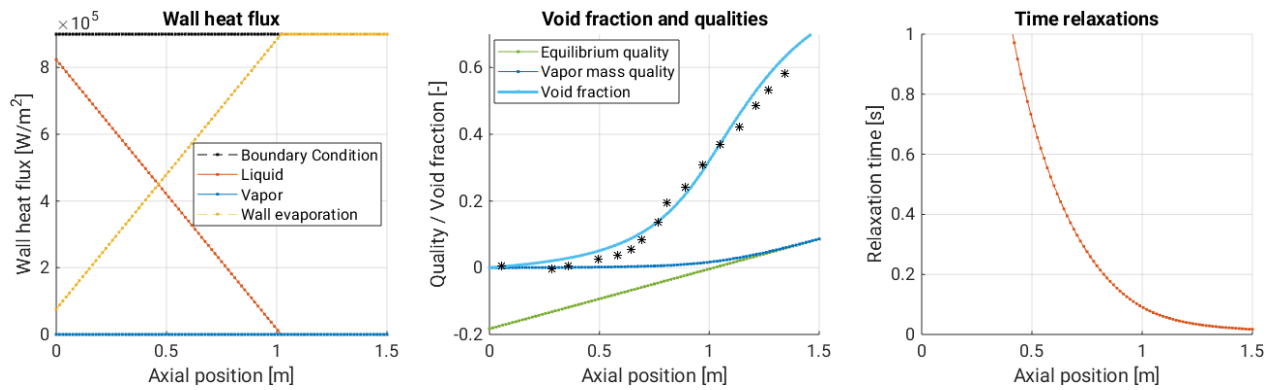


Figure 1: Axial distributions of heat flux, qualities/void fraction and interfacial condensation relaxation time for subcooled flow boiling Test #1

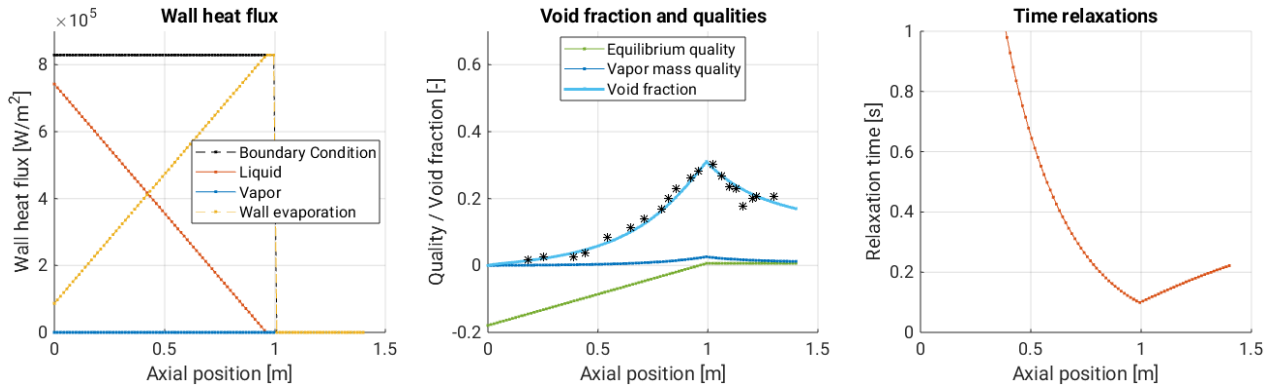


Figure 2: Axial distributions of heat flux, qualities/void fraction and interfacial condensation relaxation time for subcooled flow boiling Test #2

The results show the remarkable capability of the MRM to predict the void fraction along the test section. In particular (1) the onset of significant void and subcooled boiling is well captured and (2) the correct void fraction decay is reproduced along the adiabatic length (Figure 2, from 1.0 to 1.4 m). Consistent with Equation (18), the relaxation time is initially very large in the subcooled region (allowing wall boiling initiation) before decreasing to small values (as x and α increases) when approaching saturation. Conversely, along the adiabatic length, the relaxation time increases again (together with x and α), as expected.

In addition, Figure 3 compares the model predictions with the two limiting cases: thermal equilibrium ($t_{relax} \rightarrow 0$) and frozen ($t_{relax} \rightarrow \infty$). As expected, the model predictions (and data) correctly lie between these two limiting assumptions.

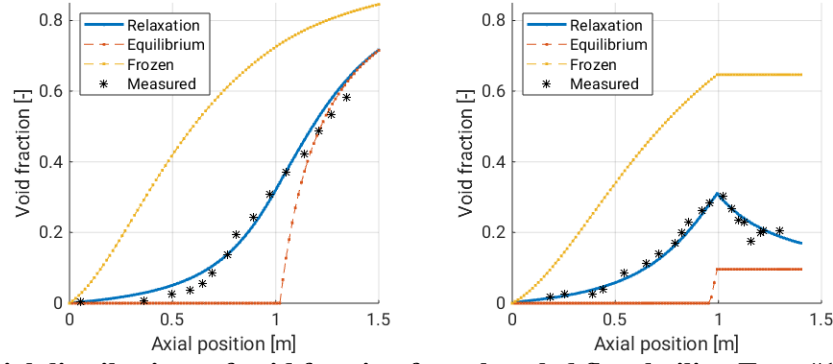


Figure 3: Axial distributions of void fraction for subcooled flow boiling Tests #1 (left) and #2 (right) for time-relaxed, thermal equilibrium and frozen models

4.2. Post-boiling transition

This section examines post-boiling transition simulations. The amount of thermal nonequilibrium in this region can be significant, as noted, e.g., in [22]. Rather than relying on a post-boiling transition correlations (e.g., [22]) to calculate the vapor mass quality ($x < x^{eq}$) and corresponding vapor temperature, or directly the wall temperature, the MRM uses the time-relaxed vapor conservation equations (Section 2.2), along with a wall heat transfer model to superheated vapor (Section 2.3.3). Note that radiation heat transfer is neglected.

Two examples from [23] have been selected to illustrate the MRM's capabilities. Wall temperature in uniformly heated convective flow boiling was measured in water at relatively high pressure (7 MPa), typical of film dryout followed by DFFB. The selected boundary conditions are listed in Table 2.

Table 2: Selected post-boiling transition test cases

Test #	Reference	Diameter [mm]	Length [m]	Pressure [MPa]	Mass flux [kg/s/m ²]	T _{inlet} [C]	Heat flux [kW/m ²]
1	[23]	14.9	7.0 (heated)	7.0	1986	275	1162
2	[23]	14.9	7.0 (heated)	7.0	501	275	511

For demonstration purposes, the same calibrated $d_{l,do}$ value (0.15 mm) was used in both cases to calculate the relaxation time related to the interfacial evaporation (according to Equation (24)). The boiling transition is imposed in the simulations at the measured elevation to focus solely on post-boiling transition heat transfer without including additional uncertainties. A homogeneous void fraction model is adopted. Figure 4 and Figure 5 show the axial distributions of relaxation time, mass quality and void fraction, vapor and wall temperatures (from Equation (11)) together with the experimental data.

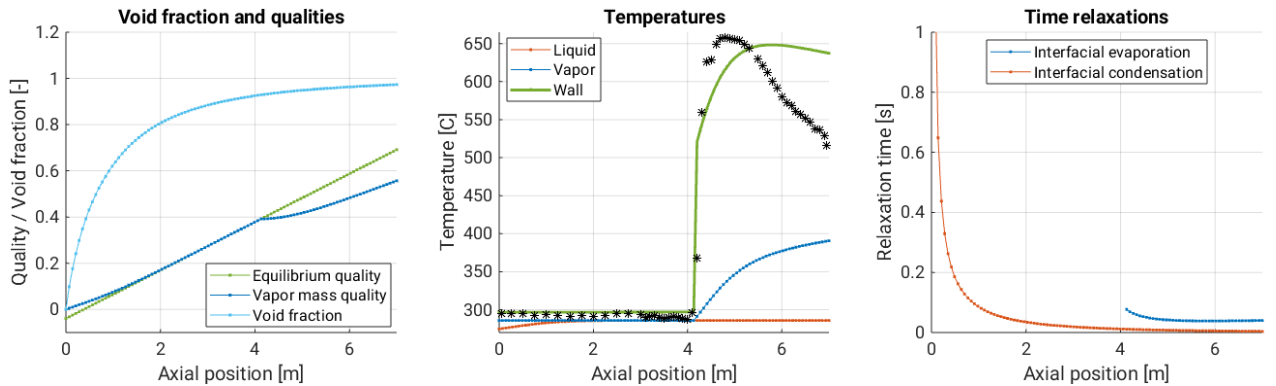


Figure 4: Axial distributions of qualities/void fraction, wall/phase temperatures and interfacial relaxation times for post-boiling transition Test #1

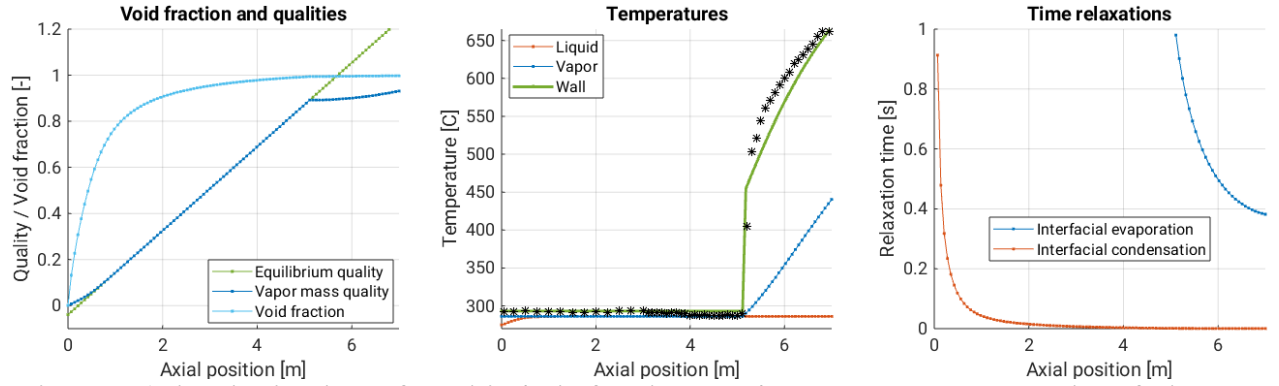


Figure 5: Axial distributions of qualities/void fraction, wall/phase temperatures and interfacial relaxation times for post-boiling transition Test #2

The results demonstrate the capability of the MRM to predict the wall temperature along the test section. In particular (1) the amplitude of the sharp wall temperature rise immediately downstream of the boiling transition is well captured (see further discussion below), and (2) the downstream behavior in the stable film boiling region is also well represented. The trend inversion observed in Test #1 due to increasing vapor Reynolds number is, however, not fully reproduced, indicating an under-estimation of interfacial evaporation. It can also be observed that the relaxation time is initially the largest at the boiling transition, decreasing toward shorter values due to the dominating effect of increased vapor thermal diffusion with temperature. The significantly lower t_{relax} for Test #1 is consistent with the higher availability of liquid (i.e., lower x) in the post-boiling region for the corresponding operating conditions.

In addition, Figure 6 compares the model predictions with the two limiting cases: thermal equilibrium ($t_{relax} \rightarrow 0$) and frozen ($t_{relax} \rightarrow \infty$). The model predictions (and data) correctly lie between these two limiting assumptions. An exception is observed for Test #2 immediately downstream the boiling transition, suggesting a slight local degradation of vapor wall heat transfer to be further investigated.

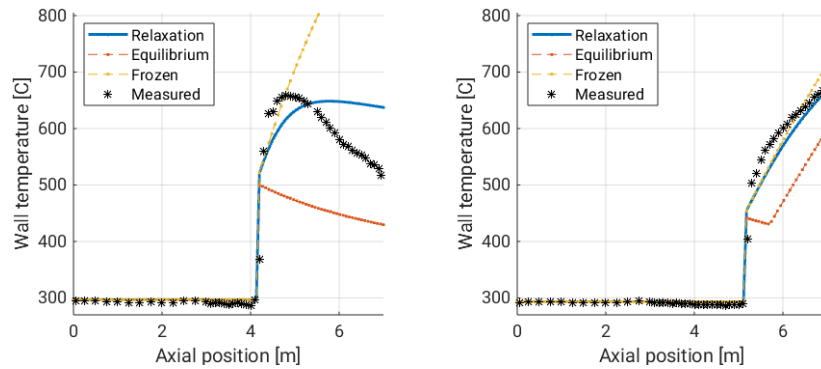


Figure 6: Axial distributions of wall temperatures for post-boiling transition Tests #1 (left) and #2 (right) for time-relaxed, thermal equilibrium and frozen models

5. CONCLUSIONS

This work introduced a Mixture Relaxation Model (MRM) implemented in the OpenSTREAM open-source code that consistently handles thermal nonequilibrium driven by wall heat flux, covering both subcooled boiling and post-boiling transition within a unified, five-equation mixture framework. The formulation permits phase slip while remaining best suited to well-coupled, high-pressure steam–water flows, supporting LWR applications such as coolant density calculations and time-at-temperature fuel safety criteria evaluations [8].

Future work will further develop physically based relaxation-time models and broaden the validation across a wider range of subcooled boiling and post-boiling transition conditions (building on recent successful performance reported in [24]), including transients.

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